

01 Observe



Local backbone conformation

ϕ , ψ ; sugar pucker and BI/BII

Crystal coordinates provide geometry

02 Correct

Fixed force field

one equilibrium per bond or angle



MechLib

equilibrium depends on conformation

03 Test

Held-out static recovery

33.1% lower RMSD • n = 300

Constrained QM • n = 80

mean favors MechLib; 3 bonds worsen

strict registered endpoint: FAIL

DNA extension: 85.4% lower median crystallographic phantom strain (descriptive)